

Python Data Structures Master Comparison

Table 1: Detailed Feature Comparison

Feature	List	Tuple	Set	Dictionary
Ordered	Yes	Yes	No	Yes (3.7+)
Mutable	Yes	No	Yes	Yes
Duplicates	Allowed	Allowed	Not Allowed	Keys Not Allowed
Indexing	Yes	Yes	No	Key Based
Slicing	Yes	Yes	No	No
Underlying Structure	Dynamic Array	Fixed Array	Hash Table	Hash Table
Search Time	O(n)	O(n)	O(1)	O(1)
Insert Middle	O(n)	Not Allowed	Not Supported	Not Supported
Hashing Used	No	No	Yes	Yes
Memory Efficient	Medium	High	Medium	Medium
Best Use	Dynamic Data	Fixed Data	Unique Values	Key-Value Mapping

Table 2: Complete Methods Comparison

Method	List	Tuple	Set	Dict
append()	Yes	No	No	No
extend()	Yes	No	No	No
insert()	Yes	No	No	No
remove()	Yes	No	Yes	No
pop()	Yes	No	Yes	Yes
clear()	Yes	No	Yes	Yes
index()	Yes	Yes	No	No
count()	Yes	Yes	No	No
sort()	Yes	No	No	No
reverse()	Yes	No	No	No
copy()	Yes	No	Yes	Yes
add()	No	No	Yes	No
update()	No	No	Yes	Yes
discard()	No	No	Yes	No

Method	List	Tuple	Set	Dict
union()	No	No	Yes	No
intersection()	No	No	Yes	No
difference()	No	No	Yes	No
symmetric_difference()	No	No	Yes	No
difference_update()	No	No	Yes	No
intersection_update()	No	No	Yes	No
symmetric_difference_update()	No	No	Yes	No
issubset()	No	No	Yes	No
issuperset()	No	No	Yes	No
isdisjoint()	No	No	Yes	No
get()	No	No	No	Yes
keys()	No	No	No	Yes
values()	No	No	No	Yes
items()	No	No	No	Yes
setdefault()	No	No	No	Yes
popitem()	No	No	No	Yes
fromkeys()	No	No	No	Yes

Table 3: Built-in Functions Comparison

Built-in Function	List	Tuple	Set	Dict
len()	Yes	Yes	Yes	Yes
max()	Yes	Yes	Yes	Keys Only
min()	Yes	Yes	Yes	Keys Only
sum()	Yes	Yes	Yes	No
sorted()	Yes	Yes	Yes	Keys Only
type()	Yes	Yes	Yes	Yes
any()	Yes	Yes	Yes	Yes
all()	Yes	Yes	Yes	Yes
list()	Yes	Yes	Yes	Yes
tuple()	Yes	Yes	Yes	Yes
set()	Yes	Yes	Yes	Yes
dict()	No	No	No	Yes
enumerate()	Yes	Yes	Yes	Yes
zip()	Yes	Yes	Yes	Yes

Built-in Function	List	Tuple	Set	Dict
reversed()	Yes	Yes	No	No
isinstance()	Yes	Yes	Yes	Yes
hash()	No	Yes	No	No
id()	Yes	Yes	Yes	Yes